

Muhammad Daffa Yunus

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Education

Universitas Gunadarma, [B.Cs](#) In Information System

Relevant Projects: AI-Powered Customer Service Chatbot

Relevant Coursework : Artificial Intelligence, Machine Learning, Natural Language Processing, Database Systems, Software Engineering, Web Programming, Information Systems Analysis and Design.

Professional Experience

Machine Learning Engineer | Project Based

April 2026 to Present

PT POWERNET INDOSOLUSI (South Jakarta, DKI Jakarta)

- Developed an AI-powered reporting platform using Large Language Models to automate report generation and improve operational efficiency.
- Integrated Retrieval-Augmented Generation pipelines to deliver accurate, personalized, and context-aware document generation from enterprise knowledge sources.
- Developed and deployed Machine Learning and Deep Learning models for predictive analytics, intelligent automation, and AI-driven business solutions.

Computer Vision Engineer | Project Based

June 2026 to Present

TRIDANA LABS (South Jakarta, DKI Jakarta)

- Developed and maintained end-to-end computer vision model pipelines, covering model loading, preprocessing, inference, and output processing for production use.
- Integrated computer vision models into existing systems and APIs, ensuring model inference could be reliably consumed by application services.
- Prepared and labeled image datasets for model development and evaluation, ensuring annotation consistency and data quality.

Software Developer | Internship

May 2026 to August 2026

DASSA CORP (Central Jakarta, DKI Jakarta)

- Developed responsive and user-friendly web application interfaces.
- Integrated RESTful APIs for dynamic data processing and presentation.
- Optimized application performance and user experience across devices.
- Built and maintained backend services and business logic.
- Managed database operations, including CRUD functionality and data integrity.
- Implemented input validation and security best practices.
- Collaborated with cross-functional teams to deliver project requirements.
- Produced technical documentation to support system development and maintenance

Project Experience

Waste Management Multi Facility Traceability Dashboard | AI Engineer

A hazardous waste traceability platform that gives companies a transparent view of the waste handling process carried out by their vendors, using verified field reporting at each stage to support B3 waste compliance, while leveraging AI-powered reporting to transform activity data into actionable environmental and sustainability insights.

- Developed and integrated AI models to support automated analysis and AI-powered reporting within the waste management platform.
- Implemented semantic key-based data selection to identify, filter, and retrieve contextually relevant information for AI processing.
- Integrated pgvector for vector storage and semantic retrieval, and Redis as a memory layer for Memory-Augmented Generation to maintain contextual information across interactions.
- Conducted model, semantic retrieval, and memory pipeline testing to evaluate response relevance, retrieval accuracy, and system reliability.
- Integrated the AI pipeline with backend services to support automated reporting and data-driven environmental insights.

License Plate Recognition | Computer Vision Engineer

A self-hosted License Plate Recognition solution developed to reducing dependency on external services and enabling a more scalable.

- Developed and evaluated a self-hosted YOLO-based License Plate Recognition model as an alternative to the existing third-party Plate Recognizer API.
- Built an end-to-end LPR pipeline combining vehicle and license plate detection, image preprocessing, PaddleOCR, and post-processing to automatically extract license plate information.
- Conducted model experiments and benchmarking across multiple YOLO configurations, evaluating detection accuracy, PaddleOCR performance, and inference speed to determine production feasibility.
- Integrated the LPR inference pipeline with FastAPI, enabling the self-hosted model to be consumed by the existing parking system through an API-based architecture.

Agentic AI Financial Advisor | AI Engineer

Agentic AI-powered financial advisor that analyzes users financial data and provides personalized insights and recommendations by dynamically selecting between user data tools and RAG-based external knowledge.

- Developed an AI Agent with tool-calling capabilities to dynamically select and execute appropriate tools based on user intent and financial context.
- Designed data tools and RAG-based knowledge retrieval tools, enabling the agent to determine when to use personal financial data or external knowledge for contextual responses.
- Implemented a Retrieval-Augmented Generation pipeline to ground financial recommendations in a curated knowledge base and reduce reliance on standalone LLM responses.
- Built the agent reasoning and response pipeline to combine retrieved knowledge, user financial data, and conversational context into personalized financial insights and recommendations.

MRV SAKALA | Software Developer

A complaint management feature within an MRV platform that enables users to submit complaints with supporting evidence and provides a structured review and approval workflow for complaint resolution.

- Developed a complaint management feature for an MRV platform, enabling users to submit, track, and manage complaints within the project workflow.

- Designed and implemented the complaint evidence workflow, allowing supporting evidence to be submitted during complaint creation and reviewed as part of the approval process.
- Developed and integrated backend APIs using AdonisJS and PostgreSQL to support complaint submission, evidence management, and approval workflows.
- Conducted API and workflow testing to validate business logic, data integrity, and end-to-end complaint processing.

TelcoSense | Machine Learning Engineer

An intelligent telco recommendation platform that combines machine learning and Agentic AI to analyze customer behavior and deliver personalized, data-driven product recommendations in real time.

- Developed and trained a Machine Learning recommendation model to analyze historical customer behavior and generate personalized telecommunication package recommendations.
- Built and optimized the ML data processing and prediction pipeline using Scikit-Learn & Numpy
- Integrated the trained recommendation model into a FastAPI microservice, enabling real-time inference and communication with the application's backend.

Achievement

2nd Winner Accounting Fest National Business Plan Competition by FEB Universitas Negeri Malang, Served as AI Engineer, developing a Retrieval-Augmented Generation system using OpenAI models to deliver context-aware, data-grounded AI responses for the proposed solution.

4th Winner Carbon Youth Challenge Business Pitch Competition by Indonesian Carbon Trade Association, Served as Engineer, developing AI-powered solutions using OpenAI models, Retrieval-Augmented Generation and data-driven approaches to support sustainability-focused solutions.

1st Winner Gunadarma Code Week 2025 National Hackathon, Served as AI Developer, developing Gemini-powered AI capabilities and a carbon emission calculator for the Slugify sustainability platform.

1st Winner Computer Multi Challenge Day National Hackathon by Universitas Syiah Kuala, Served as AI & ML Developer, developing an AI-based assessment system using supervised learning and primary data, implementing a Random Forest classifier and deploying the model through Flask, while also developing an AI role-play feature to support students' soft-skill development and contributing to model fine-tuning.

1st Winner GDGoC Bootcamp Kilat Web Development Best Project, Served as Backend Developer, contributing to the development of Candi, a warehouse management platform, and delivering the project within an intensive seven-day development period.

Technical Skills

Artificial Intelligence: Agents AI, Retrieval-Augmented Generation, Memory-Augmented Generation, Prompt Engineering, Semantic Retrieval, LLM Integration

Machine Learning: Supervised Learning, Classification, Random Forest, Model Training, Model Evaluation, Feature Engineering, Data Preprocessing

Computer Vision: YOLO, Object Detection, Optical Character Recognition, Image Preprocessing, Model Inference

Database & Vector Search : PostgreSQL, pgvector, Redis, Vector Embeddings, Semantic Search, Contextual Memory

AI & ML Frameworks: OpenAI API, TensorFlow, Scikit-learn, Pandas, NumPy, MLFlow

Backend & API Development: Python, FastAPI, Flask, AI Pipeline Integration

DevOps & Tools: Docker, Git, GitHub, Postman, Google Colab, Ngrok

Programming language : Python, JavaScript, Typescript